

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **JAIVANT.S (192311326)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **JAIVANT.S**

Assessment Tasks

Scenario-Based Questions

1. A university plans to virtualize its computer laboratory so that multiple operating systems can be provided on demand. Recommend a virtualization strategy, identify the suitable hypervisor type, and justify the decision.
2. A hospital cloud is facing VM escape and unauthorized snapshot access. Identify the major security risks and propose mitigation controls.
3. A retail company wants to modernize its datacenter using Software Defined Datacenter principles. Explain how compute, storage, and network virtualization would be introduced in phases.
4. Two cloud providers want to interoperate through federation. Explain how trust, identity, and privacy issues should be handled in the federation design.
5. An enterprise observes poor performance in a virtualized environment after overcommitting CPU and memory to multiple VMs. Analyze the scenario and recommend tuning actions.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues

Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

ANSWERS

1. University Computer Laboratory – Virtualization Strategy

The university should use **server-based virtualization** where multiple virtual machines (VMs) run on powerful physical servers.

Recommended strategy:

- Create separate VMs for Windows, Linux, and other required operating systems.
- Provide students access to VMs through the campus network.
- Use VM templates to quickly create or reset laboratory environments.
- Allocate CPU, RAM, and storage based on student requirements.

Suitable hypervisor: Type-1 (bare-metal) hypervisor

Examples include VMware ESXi, Microsoft Hyper-V, and KVM.

Justification:

- Runs directly on physical hardware.
- Provides better performance and security.
- Supports many VMs simultaneously.
- Easier to centrally manage.
- Suitable for a university lab with many users.

Conclusion: A Type-1 hypervisor with centralized VM management is the best choice because it provides flexible, scalable, and efficient access to multiple operating systems.

2. Hospital Cloud – VM Escape and Snapshot Security

The hospital faces two major security risks:

VM Escape

VM escape occurs when an attacker breaks out of a VM and gains access to the **host system or other VMs**.

Risks:

- Access to other hospital applications.
- Theft or modification of patient data.
- Compromise of the virtualization host.

Mitigation:

- Keep hypervisors and VM tools regularly patched.
- Use strong isolation between VMs.
- Apply network segmentation and firewalls.
- Use intrusion detection and monitoring.
- Restrict administrative privileges.

Unauthorized Snapshot Access

Snapshots may contain complete VM data, including sensitive patient information.

Risks:

- Exposure of patient records.
- Unauthorized copying or restoration of sensitive data.
- Data leakage through improperly stored snapshots.

Mitigation:

- Encrypt snapshots.
 - Apply strict RBAC and MFA.
 - Restrict who can create, copy, and restore snapshots.
 - Maintain audit logs.
 - Automatically delete unnecessary snapshots.
 - Store backups in secure, access-controlled locations.
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3. Retail Company – Software-Defined Datacenter Phases

The company can introduce SDDC gradually in three major phases.

Phase 1 – Compute Virtualization

- Convert physical servers into virtual machines.
- Deploy a Type-1 hypervisor.
- Create VM templates.
- Implement automated VM provisioning.
- Monitor CPU and memory utilization.

Result: Better server utilization and reduced hardware requirements.

Phase 2 – Storage Virtualization

- Combine physical storage devices into a centralized storage pool.
- Use technologies such as Software-Defined Storage (SDS).
- Allocate storage dynamically to VMs.
- Implement replication and backup.

Result: Flexible storage allocation, easier management, and improved availability.

Phase 3 – Network Virtualization

- Virtualize switches, routers, and firewalls.
- Create software-defined networks.
- Use network segmentation and virtual networks.
- Automate network configuration.

Result: Faster network provisioning, improved security, and easier scaling.

Overall: Compute → Storage → Network virtualization gradually transforms the traditional datacenter into a software-controlled infrastructure.

4. Cloud Federation – Trust, Identity and Privacy

When two cloud providers federate, they must establish clear rules for authentication, authorization, and data protection.

Trust

- Establish a formal trust relationship between providers.
- Use digital certificates and trusted Certificate Authorities.
- Define security policies and responsibilities.
- Regularly audit the federation.

Identity

- Use **Federated Identity Management**.
- Support standards such as **SAML, OAuth 2.0, or OpenID Connect**.
- Provide Single Sign-On (SSO).
- Use MFA for sensitive operations.
- Apply role-based access control.

Privacy

- Encrypt data during transmission and storage.
- Share only the minimum required user information.
- Define where data can be stored and processed.
- Maintain audit logs.
- Follow applicable privacy and data-protection regulations.
- Provide mechanisms to revoke access and delete data when required.

Example: A user authenticated by Cloud Provider A can securely access an authorized service in Cloud Provider B without creating a separate account, while Provider B receives only the identity information it needs.

5. Poor Performance Due to CPU and Memory Overcommitment

Overcommitment occurs when more virtual CPU or memory resources are assigned to VMs than the physical server can efficiently provide.

Problems

CPU overcommitment:

- High CPU ready/wait time.
- VMs compete for processor resources.
- Applications become slow.

Memory overcommitment:

- High memory pressure.
- Hypervisor may use memory reclamation techniques such as ballooning or swapping.
- Disk swapping can significantly reduce performance.

Recommended tuning actions

1. **Reduce CPU overcommitment** by removing unnecessary vCPUs from VMs.
2. **Reduce memory overcommitment** and allocate realistic RAM values.
3. Identify idle or oversized VMs and reduce their resource allocation.
4. Monitor CPU utilization, memory usage, CPU ready time, and swapping.
5. Use **VM reservations** for critical applications.
6. Apply resource limits and priorities where appropriate.
7. Migrate VMs to less-loaded hosts using **live migration**.
8. Add physical CPU/RAM capacity if demand consistently exceeds available resources.

Conclusion: The company should first identify oversized VMs and reduce unnecessary allocations, then rebalance workloads across hosts. Hardware should be added if sustained demand remains higher than the available capacity.
